

THE SCOTS COLLEGE
Sydney

2004
TRIAL H.S.C.
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 120

- Attempt Questions 1-10
- All questions are of equal value
- Start a new booklet for every question

STUDENTS ARE ADVISED THAT THIS IS A TRIAL EXAMINATION ONLY AND CANNOT IN ANY WAY GUARANTEE THE CONTENT OR THE FORMAT OF THE HIGHER SCHOOL CERTIFICATE EXAMINATION.

QUESTION ONE

[12 MARKS]

- a. Evaluate correct to 1 decimal place $\frac{m^2 + n^2}{mn}$, where $m = -4.3$ and $n = 2.1$ 2
- b. A television is bought at a 20% discount sale for \$760. Calculate the original price of the television. 2
- c. Find the primitive function for $5 - \sqrt{x}$ 2
- d. Solve $|x - 7| > 2$ 2
- e. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$ 2
- f. Simplify fully $\frac{m^2}{m^2 + 3m + 2} - \frac{2m}{m + 2}$ 2

QUESTION TWO**[12 MARKS]**

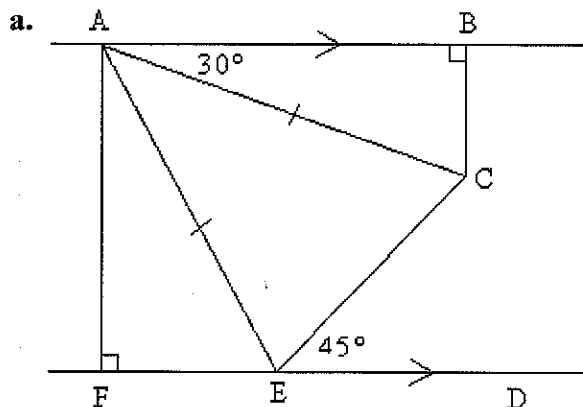
- a. Let α and β be the roots of the equation $x^2 - 7x + 3 = 0$. Calculate:
- i. $\alpha + \beta$ 1
 - ii. $\alpha\beta$ 1
 - iii. $\alpha^2 + \beta^2$ 1
 - iv. $\frac{1}{\alpha} + \frac{1}{\beta}$ 1
- b. P is the point (4,-8) and l is the line with equation $5x - 12y + 53 = 0$.
- i. Calculate the gradient of the line l and the angle that it makes with the x axis. (Answer to the nearest degree.) 2
 - ii. Show that the perpendicular to l through P has equation $12x + 5y - 8 = 0$. 2
 - iii. Find the point of intersection, Q, of this perpendicular line with the line l . 2
 - iv. Find the distance PQ. 2

QUESTION THREE**[12 MARKS]**

- a. Differentiate with respect to x :
- i. $5 \sin^2 x$ 2
 - ii. $\ln(4x^3 + 3x)$ 2
 - iii. $e^{(8x^3 - 5x)}$ 2
- b. Find:
- i. $\int (3x + 4)^5 dx$ 1
 - ii. $\int_1^{e^4} \frac{6}{x} dx$ 2
- c.
- i. Write down the discriminant of $5x^2 + 3x + k$. 1
 - ii. For what values of k does $5x^2 + 3x + k = 0$ have real roots? 2

QUESTION FOUR

[12 MARKS]

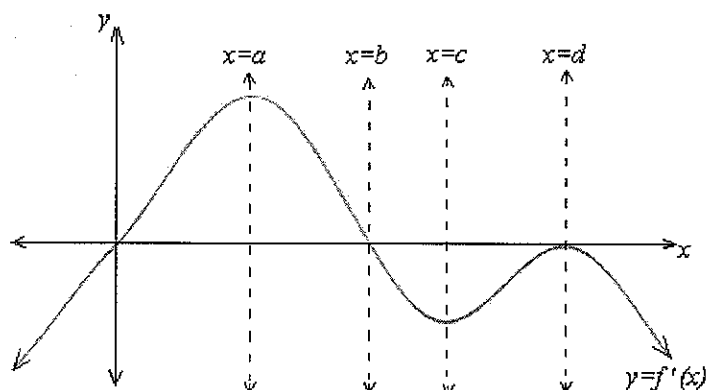


In the diagram $AC = AE$ and AB is parallel to FD , angle $CED = 45^\circ$ and angle $BAC = 30^\circ$, BC is perpendicular to AB and AF is perpendicular to FD .

Copy this diagram into your answer booklet.

- i. Find the size of angle ACE giving reasons. 2
- ii. Hence find the size of angle CAE giving reasons. 2
- iii. Prove $\triangle ABC \equiv \triangle AFE$. 3

b. Below is the graph of $y = f'(x)$



- i. For the graph of $y = f(x)$, what would be the feature that would occur when:
 - α) $x = b$ 1
 - β) $x = c$ 1
 - γ) $x = d$ 1
- ii. Transfer this sketch into your answer booklet and sketch a graph of $y = f''(x)$ onto the same axes. 2

QUESTION FIVE**[12 MARKS]**

- a. A company making shoes makes 200 pairs in the first month of operation. They intend to increase their output by 25 pairs a month. How many pairs of shoes do they intend to make:
- i. in their 24th month of operation; 2
 - ii. in total over the two year period. 2
- b. Describe the graph of $x^2 + y^2 + 4x - 6y - 3 = 0$ as accurately as possible. 2
- c. A bag contains 10 blue marbles, 5 red marbles and 1 green marble. Tim selects one marble and does not replace it before drawing a second marble.
- i. Draw a probability tree to show all of the possible outcomes. 2
 - ii. Find the probability that both of the marbles are blue. 1
 - iii. Find the probability that at least one of the marbles is blue. 2
 - iv. Find the probability that the two marbles are the same colour. 1

QUESTION SIX**[12 MARKS]**

- a. Consider the two curves $y + x^2 - 6 = 0$ and $y + 2x - 3 = 0$.
- i. Find the points of intersection. 2
 - ii. Calculate the area between the two curves. 2
- b. A geometric series has the seventh term 640 and fourth term 80. Find the first term and common ratio. 3
- c. Solve $2 \cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. 2
- d. Sketch the graph of the function $y = 10 - 5 \sin 2x$ for $0 \leq x \leq \pi$. 3

QUESTION SEVEN [12 MARKS]

- a. Consider the curve $y = 4x^3 - 6x^2 - 24x + 1$.
- i. Find the stationary points and determine their nature. 3
 - ii. Find any points of inflexion. 2
 - iii. Sketch the curve for the domain $-2 \leq x \leq 3$. 2
 - iv. For what values of x is the curve increasing? 1
- b. Consider the parabola $x^2 - 14x - 8y + 41 = 0$.
- i. By first rewriting the function in the form $(x - b)^2 = 4a(y - c)$, find the coordinates of the vertex. 2
 - ii. Find the focal length and the coordinates of the focus of this parabola. 2

QUESTION EIGHT [12 MARKS]

- a. The population of Scotsville at the beginning of 1980 was 12000 and at the beginning of 1990 it was 13080. Assume the population of Scotsville is governed by the equation $P = P_0 e^{kt}$ where t is in years, P_0 and k are both constants and P is the population at time t .
- i. Find k correct to 4 decimal places. 2
 - ii. What would the population be at the start of 2010? 2
 - iii. In which year would the population reach 20000? 2
- b. The velocity of a particle moving along the x axis is given by $v = 8t - t^2$.
- i. When will the particle be at rest? 1
 - ii. If the particle was originally 3 metres to the right of the origin, find the displacement of the particle (x) as a function of time (t). 2
 - iii. When will the particle have maximum velocity? 1
 - iv. What is the total distance travelled by the particle in the first 9 seconds? 2

QUESTION NINE**[12 MARKS]**

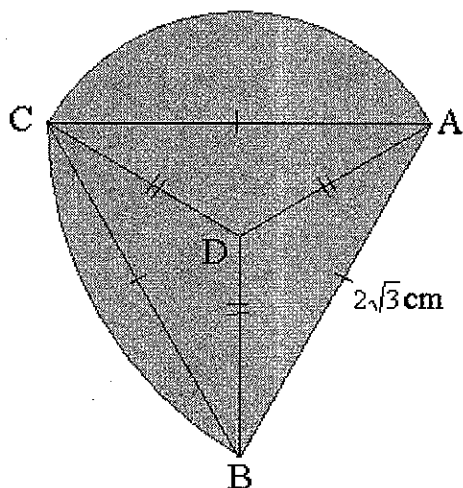
- a. Find the exact volume of revolution when the area bounded by the curve $y = 3 \sec x$ and the x axis between $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x axis. 2
- b. Consider the function $f(x) = 6 \log_e \left(\frac{x-3}{5} \right)$
- i. What is the domain of the function $y = f(x)$ 1
 - ii. Sketch the curve $y = f(x)$ showing all important features. 2
 - iii. Use Simpson's Rule with 5 ordinate values to find an approximation for the area between the curve, the x axis and the ordinates $x = 4$ and $x = 8$. (Answer to 3 d.p.) 4
- c. Solve the equation $\ln(2x+3) + \ln 4 = 2 \ln 2x$ for x . 3

QUESTION TEN**[12 MARKS]**

- a. A 12m long piece of string is cut into two pieces to form a circle of radius r metres and a square of side x metres.
- i. Show that the total area of the circle and square is given by 3

$$A = \pi r^2 + 9 - 3\pi r + \frac{\pi^2 r^2}{4}$$
 - ii. Find the exact radius r of the circle such that the total area will be a minimum. 4

b.



The 2 dimensional shape shown has the arc BC formed from centre A and the arc CA formed from centre D. AC, BC and AB are all equal. A, C and B are all equidistant from D. AB is $2\sqrt{3}$ cm. Calculate the exact area of this shape. 5

END OF EXAMINATION

THE SCOTS COLLEGE

2 Unit Trial HSC 2004

QUESTION ONE [12 MARKS]

a.
$$\frac{m^2 + n^2}{mn} = \frac{(-4.3)^2 + (2.1)^2}{-4.3 \times 2.1}$$

$$= -2.5 \text{ (to 1 d.p.)}$$

b. $80\% = \$760$
 $1\% = \$9.50$
 $100\% = \$950$

c. primitive of $5 - x^{1/2}$

$$= 5x - \frac{2x^{3/2}}{3} + c$$

$$= 5x - \frac{2\sqrt{x^3}}{3} + c$$

d. $|x - 7| > 2$
 $-2 > x - 7 \text{ and } x - 7 > 2$
 $5 > x \text{ and } x > 9$

e.
$$\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 1)}{x - 3}$$

$$= \lim_{x \rightarrow 3} (x + 1)$$

$$= 4$$

f.
$$\frac{m^2}{m^2 + 3m + 2} - \frac{2m}{m + 2}$$

$$= \frac{m^2}{(m + 2)(m + 1)} - \frac{2m}{m + 2}$$

$$= \frac{m^2 - 2m(m + 1)}{(m + 2)(m + 1)}$$

$$= \frac{m^2 - 2m^2 - 2m}{(m + 2)(m + 1)}$$

$$= \frac{-m^2 - 2m}{(m + 2)(m + 1)}$$

$$= \frac{-m(m + 2)}{(m + 2)(m + 1)}$$

$$= \frac{-m}{m + 1}$$

QUESTION TWO [12 MARKS]

a. i. $\alpha + \beta = 7$

ii. $\alpha\beta = 3$

iii. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= 49 - 2(3)$$

$$= 43$$

iv.
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$$

$$= \frac{-7}{3}$$

b.i. $5x - 12y + 53 = 0$
 $12y = 5x + 53$
 $y = \frac{5}{12}x + \frac{53}{12}$
 Gradient = $5/12$
 Angle $\tan \theta = \frac{5}{12}$
 $\theta = 23^\circ$

ii. $m = -12/5$ thru $(4, -8)$
 $y - (-8) = -\frac{12}{5}(x - 4)$
 $5y + 40 = -12x + 48$
 $12x + 5y - 8 = 0.$

iii. $12x + 5y - 8 = 0 \text{ ---(1)}$
 $5x - 12y + 53 = 0 \text{ --(2)}$
 $(1) \times 5 \text{ and } (2) \times 12$
 $60x + 25y - 40 = 0 \text{ ---(3)}$
 $60x - 144y + 636 = 0 \text{ --(4)}$
 $(4) - (3)$
 $-169y + 676 = 0$
 So $y = 4$
 Subst. into (1) $12x + 20 - 8 = 0$
 $x = -1$ therefore intersection $(-1, 4)$

iv. $PQ = \sqrt{(4 - -8)^2 + (-1 - 4)^2}$
 $PQ = \sqrt{144 + 25}$
 $PQ = 13$

QUESTION THREE

[12 MARKS]

a.i. $\frac{d5\sin^2 x}{dx}$
 $= \frac{d5(\sin x)^2}{dx}$
 $= 10 \cos x \cdot \sin x$

ii. $\frac{d \log_e (4x^3 + 3x)}{dx}$
 $= \frac{12x^2 + 3}{4x^3 + 3x}$

iii. $\frac{de^{(8x^3-5x)}}{dx} = (24x^2 - 5)e^{(8x^3-5x)}$

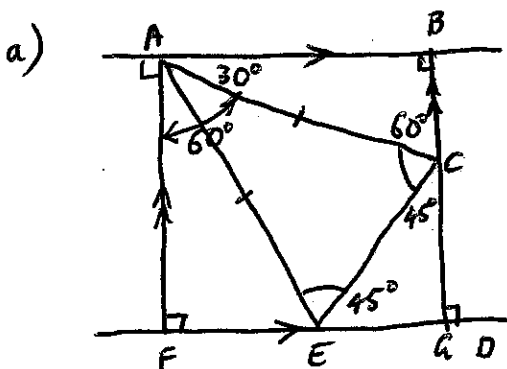
b. i. $\int (3x+4)^5 dx$
 $= \frac{1}{18} (3x+4)^6 + c$

ii. $\int_{e^4}^1 \frac{6}{x} dx = [6 \ln x]_{e^4}^1$
 $= [6 \ln e^4 - 6 \ln 1]$
 $= 24$

c.i. $5x^2 + 3x + k$
 $\Delta = 9 - 4 \cdot 5 \cdot k$
 $\Delta = 9 - 20k$

ii. real roots $\Delta \geq 0$
 $9 - 20k \geq 0$
 $9 \geq 20k$
 $\frac{9}{20} \geq k$

Question Four

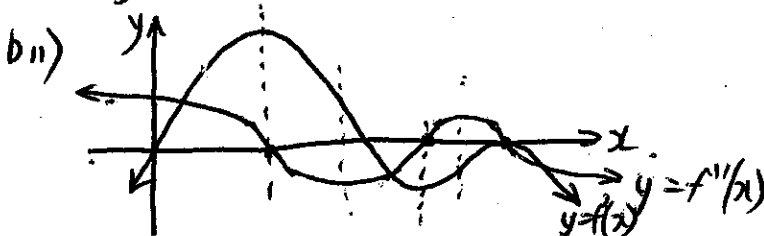


i) Produce BC to the line FD
 call the point G.
 $\angle ACB + \angle ACE + \angle EGH = 180$
 $60^\circ + \angle ACE + 45 = 180$
 $\angle ACE = 180 - 60 - 45$
 $= 75^\circ$

ii) $\triangle ACE$ is isosceles
 $\therefore \angle AEC = 75^\circ$
 $\therefore \angle CAE = 30^\circ$ (angles sum of $\triangle$)

iii) In $\triangle$'s ABC and AFE
 $AC = AE$ (given). S
 $\angle ABC = \angle AFE$ (90° given perpendicular)
 $\angle PAE = 90 - \angle BAC - \angle CAE$
 so $\angle PAE = 30^\circ$ (angles adding to 90°).
 Now $\angle BAC = \angle PAE$ A.
 $\therefore \triangle ABC \equiv \triangle AFE$ (AAS)

b) i) $f'(b) = 0 \therefore$ turn pt.
 $f'(a^-) = +ve$ $f'(a^+) = -ve$.
 $\therefore$ max. turn point.
 B) $f'(c)$ minimum
 $\therefore$ pt. of inflexion
 3) $f'(d) = 0$ also gradient of $f'(x) = 0$.
 so $f'(d) = 0$ and $f''(d) = 0$
 so horizontal point of inflexion.

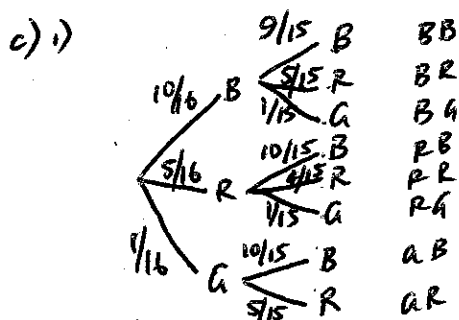


Question Five

a) A.P.
 $a = 200$ $d = 25$.
 $T_{24} = 200 + 23 \times 25$
 $= 200 + 575$
 $= 775$.

ii) $S_{24} = \frac{n}{2}(a + l)$
 $= \frac{24}{2}(200 + 775)$
 $= 12(975)$
 $= 11700$

b) $x^2 + y^2 + 4x - 6y - 3 = 0$
 $x^2 + 4x + y^2 - 6y = 3$
 $(x+2)^2 - 4 + (y-3)^2 - 9 = 3$
 $(x+2)^2 + (y-3)^2 = 16$
 CIRCLE Centre $(-2, 3)$ radius 4



ii) $P(B\bar{B}) = \frac{10}{16} \times \frac{9}{15}$
 $= \frac{3}{8}$

iii) $P(\text{at least 1 blue})$
 $= P(B\bar{B}) + P(\bar{B} \cap B) + P(BB)$
 $= \left(\frac{10}{16} \times \frac{9}{15}\right) + \left(\frac{6}{16} \times \frac{10}{15}\right) + \frac{3}{8}$
 $= \frac{1}{4} + \frac{1}{4} + \frac{3}{8}$
 $= \frac{7}{8}$

iv) $= P(BB) + P(RR) + P(RG)$
 $= \left(\frac{10}{16} \times \frac{9}{15}\right) + \left(\frac{6}{16} \times \frac{4}{15}\right)$
 $= \frac{3}{8} + \frac{1}{12}$
 $= \frac{11}{24}$

QUESTION SIX

a) $y = 6 - x^2$ — (1)
 $y = -2x + 3$ — (2)

from (1) and (2)
 $6 - x^2 = -2x + 3$
 $x^2 - 2x - 3 = 0$
 $(x-3)(x+1) = 0$
 $x = 3$ and $x = -1$.
 when $x = 3$ $x = -1$
 $y = 6 - 9$ $y = 6 - 1$
 $= -3$ $= 5$.

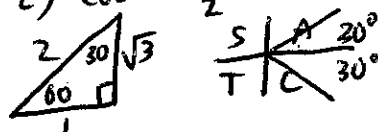
ii) $A = \int_{-1}^3 (6 - x^2) - (-2x + 3) dx$
 $= \int_{-1}^3 6 - x^2 + 2x - 3 dx$
 $= \int_{-1}^3 -x^2 + 2x + 3 dx$
 $= \left[-\frac{x^3}{3} + x^2 + 3x \right]_{-1}^3$
 $= (-9 + 9 + 9) - \left(-\frac{1}{3} + 1 - 3 \right)$
 $= 10\frac{2}{3} \text{ units}^2$

b) G.P. $T_7 = 640$ $T_3 = 80$
 $ar^6 = 640$ — (1)
 $ar^2 = 80$ — (2).

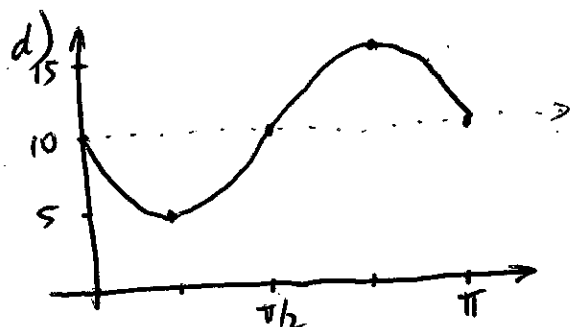
(1) $\div$ (2)
 $r^4 = 8$ so $r = 2$
 subst. into (2)

a) $2^3 = 80$
 $a = 10$

c) $\cos x = \frac{\sqrt{3}}{2}$ $x = 30^\circ$.



$x = 30^\circ$ and 330°



QUESTION SEVEN

$$y = 4x^3 - 6x^2 - 24x + 1$$

$$y' = 12x^2 - 12x - 24$$

$$y'' = 24x - 12$$

i) stat pts $y' = 0$

$$12x^2 - 12x - 24 = 0$$

$$12(x^2 - x - 2) = 0$$

$$(x-2)(x+1) = 0$$

so $x = 2$ and $x = -1$.

when $x = -1$ $x = 2$

$$y = 15$$

$$y = -39$$

$$y'' = -ve$$

$$y'' = +ve$$

so max turnpt. so min turnpt.

at $(-1, 15)$

at $(2, -39)$

ii) inflexion when $y'' = 0$

$$24x - 12 = 0$$

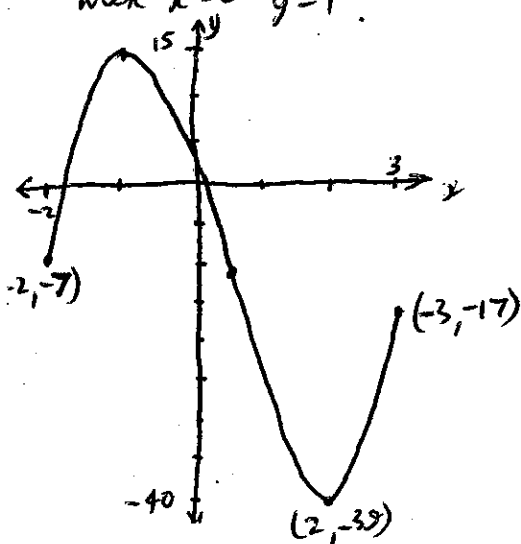
$$x = 1/2$$

x	$1/2^-$	$1/2$	$1/2^+$
$f'''(x)$	-	0	+

when $x = 1/2$ $y = -12$

iii) end points $x = -2$ $y = -7$
 $x = 3$ $y = -17$

when $x = 0$ $y = 1$



$$b) x^2 - 14x - 8y = -49$$

$$(x-7)^2 - 49 - 8y = -49$$

$$(x-7)^2 = 8y + 8$$

$$(x-7)^2 = 8(y+1)$$

$$v(7, -1) a = 2$$

ii) focal length = 2

Focus

$$= (7, 1)$$

QUESTION EIGHT

a) let 1980 be $t = 0$

$$\text{so } P_0 = 12000$$

ii) when $t = 10$ $P = 13080$

$$\text{So } 13080 = 12000 e^{10k}$$

$$e^{10k} = \frac{13080}{12000}$$

$$k = \left(\ln \frac{1308}{1200} \right) \div 10$$

$$= 0.0086 \text{ to 4 d.p.}$$

ii) when $t = 30$

$$P = 12000 e^{30k}$$

$$= 15540$$

iii) find t when $P = 20000$

$$20000 = 12000 e^{0.0086t}$$

$$e^{0.0086t} = \frac{20000}{12000}$$

$$t = \ln(5/3) \div 0.0086$$

$$= 57.3 \text{ years}$$

so in 60th year i.e 2040

b) $v = 8t - t^2$

i) rest means $v = 0$

$$8t - t^2 = 0$$

$$t(8-t) = 0$$

$$\text{so } t = 0 \text{ and } t = 8$$

$$ii) x = 4t^2 - \frac{t^3}{3} + c$$

$$\text{when } t = 0, x = 3$$

$$\text{so } c = 3$$

$$\therefore x = 4t^2 - \frac{t^3}{3} + 3$$

iii) Max vel. when $\dot{x} = 0$

$$\frac{dv}{dt} = 8 - 2t$$

so when $t = 4$ max. will occur as $8t - t^2$ is concave down

iv)

no change of direction when $t = 8$.

when

$$t = 0, x = 3$$

$$t = 8, x = 88\frac{1}{3}$$

$$t = 9, x = 84$$

$$\text{First 8 seconds distance} = 85\frac{1}{3} \text{ m}$$

$$\text{last second} = 9\frac{1}{3} \text{ m}$$

$$\text{TOTAL} = 89\frac{2}{3} \text{ m}$$

QUESTION NINE

a) $y = 3 \sec x$
 $y^2 = 9 \sec^2 x$

so $V = \pi \int_0^{\pi/4} 9 \sec^2 x dx$

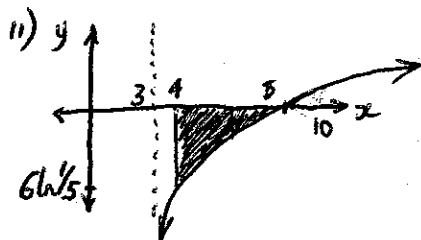
$= 9\pi [\tan x]_0^{\pi/4}$

$= 9\pi \text{ units}^3$

b) $f(x) = 6 \log_e \left(\frac{x-3}{5} \right)$

i) domain $\frac{x-3}{5} > 0$

$x-3 > 0$
 $x > 3$



iii)

x	4	5	6	7	8
f(x)	-9.66	6 ln 0.4	6 ln 0.6	6 ln 0.8	0

$A = \frac{1}{3} \left[f(4) + f(8) + 2 \left\{ f(5) + f(7) \right\} + 4 \left\{ f(6) \right\} \right]$
 $= \frac{1}{3} \left[-9.66 + 0 + 4(6 \ln 0.4 + 6 \ln 0.8) + 2(6 \ln 0.6) \right]$

≈ 14.378 note all below axis.

c) $\ln(2x+3) + \ln 4 = 2 \ln 2x$

$\ln[(2x+3) \times 4] = \ln(2x)^2$

$8x+12 = 4x^2$

$4x^2 - 8x - 12 = 0$

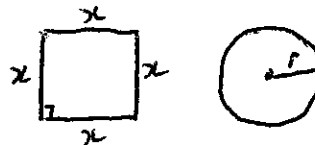
$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$x = 3 \text{ and } x = -1$

QUESTION TEN

a)



$P = 4x$

$A = x^2$

$C = 2\pi r$

$A = \pi r^2$

total perimeter $= 4x + 2\pi r$
 $= 12$

$4x + 2\pi r = 12$

$4x = 12 - 2\pi r$

$x = 3 - \frac{\pi r}{2}$

so $A = x^2 + \pi r^2$

$= \left(3 - \frac{\pi r}{2} \right)^2 + \pi r^2$

$= 9 - 3\pi r + \frac{\pi^2 r^2}{4} + \pi r^2$

$= \pi r^2 + 9 - 3\pi r + \frac{\pi^2 r^2}{4}$

ii) $A = \pi r^2 + 9 - 3\pi r + \frac{\pi^2 r^2}{4}$

$A' = 2\pi r - 3\pi + \frac{\pi^2 r}{2}$

$A'' = 2\pi + \frac{\pi^2}{2}$

Max and min area when $A' = 0$

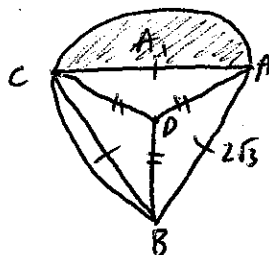
$r \left(2\pi + \frac{\pi^2}{2} \right) - 3\pi = 0$

$r = \frac{3\pi}{2\pi + \pi^2/2}$

$= \frac{6\pi}{4\pi + \pi^2}$

$r = \frac{6}{4 + \pi}$

this is a minimum as A'' is +ve.



Area =

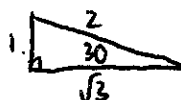
$\frac{1}{2} \times 2 \times 2 \times \sin 120^\circ + \frac{1}{6} \times \pi \times (2\sqrt{3})^2$

$= \frac{4\pi}{3} - \sqrt{3} + 2\pi$

$= \frac{10\pi}{3} - \sqrt{3} \text{ cm}^2$

$\angle AOB = 120^\circ$

$\angle OAB = 30^\circ$



$\therefore AD = 2$